

SilentSOS ? Project Technology Audit Report

Google Maps Restricted Strictly to Map/Location Visualization

Google APIs Cleanup Complete | Google Maps Restricted Strictly to Location Visualization

1 Frontend Technology Stack & Architecture

Framework / Core:	React 18.3.1 (Single Page App)	Programming Language:	TypeScript 5.5.4 / JavaScript ES Modules
Build Tool & Bundler:	Vite 5.2.0 + @vitejs/plugin-react	CSS / UI Framework:	Tailwind CSS 3.4.17 + PostCSS + Autoprefixer
State Management:	React Context API (AppContext.tsx)	Client-Side Routing:	React Router DOM 6.26.2 (Role-Guarded)
Animation Engine:	Framer Motion 11.5.4	Iconography:	Lucide React 0.522.0
Mapping Libraries:	Google Maps JS API (Location Only) + Leaflet 1.9.0	Computer Vision / AI:	TensorFlow.js 4.22.0 + MediaPipe Models
Date Utility:	date-fns 4.0.0	CSS-in-JS Engine:	@emotion/react 11.13.3

2 Backend Engine, APIs & Security

Backend Runtime:	Node.js (Native ES Modules)	Web Framework:	Express 4.19.2 + Native HTTP
Real-time Protocol:	WebSocket (ws 8.21.3) for GPS & Alert Stream	Server Entry Point:	backend/server.js (with db.js)
Authentication:	Custom Base64 Bearer Token Middleware	Password Hashing:	SHA-256 (Node.js crypto.createHash)
Data Encryption:	AES-256-CBC (Field level for GPS/Contacts)	Media Upload Storage:	Multer 1.4.5 (Multipart disk storage)

Method	Endpoint	Auth Level	Purpose / Function
POST	/api/auth/register	Public	User registration and account creation
POST	/api/auth/login	Public	User authentication and token issue
POST	/api/auth/forgot-password	Public	Dispatches 6-digit email reset code
POST	/api/auth/reset-password	Public	Verifies OTP and resets user password
GET	/api/alerts/:id	Public	Live receiver coordinate tracker data
POST	/api/alerts/:id/evidence	Public/ID	Uploads photos, audio & video evidence
GET	/api/state	Bearer Token	Hydrates complete user profile and alerts
POST	/api/alerts	Bearer Token	Triggers live SOS alert & emergency dispatch
POST	/api/alerts/:id/cancel	Bearer Token	Cancels alert & sends resolved notices
GET	/api/admin/stats	Admin Token	Platform analytics and alert statistics
GET	/api/admin/alerts/export	Admin Token	Exports all distress records to CSV

3 Database Architecture & Schema Analysis

Database Engine:	PostgreSQL (Single source of truth)	Cloud Provider:	Neon Serverless PostgreSQL (port 443 pooling)
SQL Drivers:	pg 8.21.0 & @neondatabase/serverless 1.1.0	Connection File:	backend/db.js (auto-adapts Neon vs standard PG)
Encrypted Columns:	phone_enc, email_enc, gps_path_enc (AES-256) JSONB Columns: contacts.preferences, history.evidence, history.contacts		

Table Name	Primary Key	Foreign Keys	Key Fields & Schema Description
users	id (VARCHAR)	None	email (UNIQUE), password_hash, name, role, disabled, is_setup_complete
contacts	id (VARCHAR)	user_id -> users(id)	name, phone_enc (AES), email_enc (AES), preferences (JSONB)
settings	user_id (VARCHAR)	user_id -> users(id)	gesture_sensitivity, auto_repeat_interval, photo_burst, safety_pin, global_emails
history	id (VARCHAR)	user_id	timestamp, type, duration_seconds, status, evidence (JSONB), gps_path_enc (AES)
password_resets	email (VARCHAR)	None	code (VARCHAR 10), expires_at (BIGINT Unix epoch timestamp)

4 Third-Party Services & Integrations

Service / API	Category	Configuration Key	Application Role
Google Maps JS API	Mapping & Geo	VITE_GOOGLE_MAPS_API_KEY	Renders live location maps, trails & dark markers ONLY
Nodemailer (Gmail SMTP)	Email Dispatch	SMTP_HOST / SMTP_PASS	Sends SOS warning emails, evidence & reset OTPs
Resend HTTP API	Cloud Email	RESEND_API_KEY	Secondary cloud REST email dispatching pipeline
Fast2SMS Gateway	Emergency SMS	FAST2SMS_API_KEY	Sends real-time SMS alerts to Indian mobile numbers
Twilio SDK	Telecom SDK	backend/package.json	Telecommunications and emergency messaging SDK
Neon Database	Serverless DB	DATABASE_URL	Cloud serverless PostgreSQL persistent storage
Browser Geolocation	Device GPS	navigator.geolocation	Obtains latitude, longitude & accuracy from device GPS
MediaDevices API	Device Camera	getUserMedia	Captures camera stream for gesture scan & evidence
TensorFlow / MediaPipe	Client-side AI	npm models	Detects palm and blink distress gestures in browser

5 Environment Variables Audit (Post-Cleanup)

Frontend: VITE_API_BASE	Present (Target API URL)	Frontend: VITE_GOOGLE_MAPS_API_KEY	Present (Google Maps JS Key)
Backend: PORT	Present (Default: 3001)	Backend: SMTP_HOST / SMTP_PASS	Present (Email / SMTP Server)
Backend: SMTP_USER / SMTP_PASS	Present (Credentials Hidden)	Backend: DATABASE_URL	Present (PostgreSQL Connection String Hidden)
Backend: FAST2SMS_API_KEY	Present (SMS Gateway API Key Hidden)	Backend: GOOGLE_MAPS_API_KEY	REMOVED (Backend does not call Google APIs)

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Location Architecture: GPS vs Google Maps

Decoupled GPS Collection & Google Maps Visualization Flow

1. DEVICE GPS ACQUISITION: Browser HTML5 Geolocation API (useGeolocation.ts) reads device GPS.
2. COORDINATE HARVESTING: Latitude, longitude, and accuracy radius update in AppContext state.
3. GOOGLE MAPS VISUALIZATION: GoogleMapTracker.tsx receives coordinates to render markers & breadcrumb trails.
4. SOS BROADCAST: Server writes encrypted GPS points to PostgreSQL and streams over WebSockets to /receiver/:id.
5. SEPARATION OF CONCERNS: Google Maps is strictly for VISUALIZATION; Device GPS collects raw coordinates.

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Complete Package & Dependency Matrix

Package Name	Version	Scope	Role & Usage
react	^18.3.1	Frontend	User interface component engine
react-router-dom	^6.26.2	Frontend	Declarative client-side routing & navigation
typescript	^5.5.4	Frontend	Type definition compiler & build validation
vite	^5.2.0	Frontend	Development server & production bundler
tailwindcss	3.4.17	Frontend	Utility-first design system & styling
framer-motion	^11.5.4	Frontend	Micro-interactions and gesture animations
lucide-react	0.522.0	Frontend	Vector icon library
@tensorflow/tfjs	^4.22.0	Frontend	Client-side neural network execution runtime
@mediapipe/hands	^0.4.x	Frontend	Hand landmark detection for palm gesture triggers
express	^4.19.2	Backend	HTTP application server framework & routing
ws	^8.21.3	Backend	WebSocket server for bidirectional live streaming
pg	^8.21.0	Backend	PostgreSQL database client and connection pool
@neondatabase/serverless	^1.1.0	Backend	Neon serverless HTTPS PostgreSQL adapter
nodemailer	^8.0.11	Backend	Automated SMTP email dispatching transport
multer	^1.4.5-lts.1	Backend	Multipart form-data & file upload middleware

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Authentication & Deployment Status

User Auth Architecture:	Token-Based (Base64 JSON Token payload)	Password Security:	SHA-256 Hashing with unique salt string
Password Reset Pipeline:	6-digit OTP verification email with 10-minute expiry	Session Storage:	Synchronized in localStorage & sessionStorage
Frontend Deployment:	Vercel (Detected via CNAME & vercel.json rewrite)	Backend Deployment:	Render (Targeted via VITE_API_BASE in .env.production)
Database Hosting:	Neon Serverless PostgreSQL Cloud Platform		

EXECUTIVE TECHNOLOGY AUDIT SUMMARY			
? Frontend:	React 18, TypeScript, Vite, Tailwind CSS, Framer Motion	? Backend:	Node.js, Express REST API, WebSocket (ws), Multer
? Database:	PostgreSQL / Neon Serverless (AES-256-CBC Encryption)	? Authentication:	Bearer Tokens, SHA-256 Hashing, 6-digit OTP Email Reset
? Google Maps API:	Maps JavaScript API ONLY (Location Visualization)	? Integrations:	Nodemailer (Gmail SMTP), Resend, Fast2SMS, Twilio, Neon
? Hosting:	Frontend: Vercel Backend: Render Database: Neon		
		? Languages:	TypeScript, JavaScript, SQL, HTML, CSS